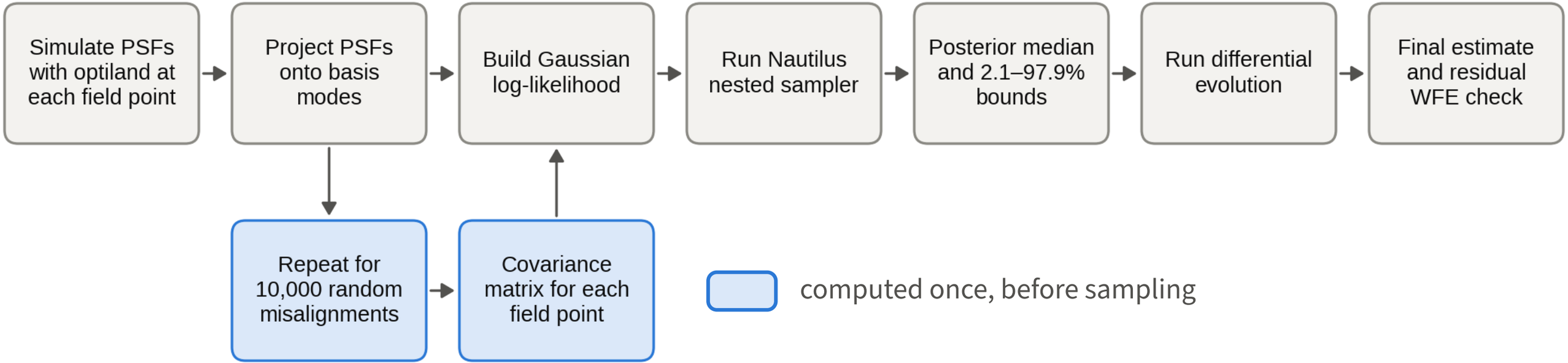

Simulate PSFs with optiland at each field point
Project PSFs onto basis modes
Build Gaussian log-likelihood
Run Nautilus nested sampler
Posterior median and 2.1–97.9% bounds
Run differential evolution
Final estimate and residual WFE check
Repeat for 10,000 random misalignments
Covariance matrix for each field point
computed once, before sampling